

Conservation dimension d_c : a controllable structural difficulty axis for LLM physics reasoning

each independent conservation law multiplies the odds of a correct answer by $OR \approx 0.69$ — a constraint-penalty law that needs no information-theoretic machinery

1 Count conservation constraints



elastic collision

momentum + energy

$$d_c = 2$$

2 Constraint-penalty law

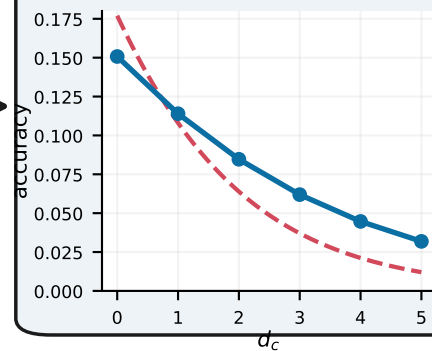
$$p \times p \times p$$

correct only if ALL pass

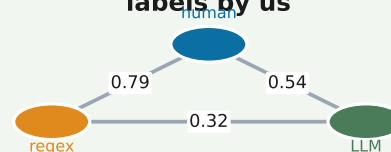
$$P = p^{d_c} \Rightarrow \text{logit } P \propto d_c$$

$OR \approx 0.69$ per constraint

3 Accuracy falls with d_c



4a Validated, no human labels by us



4b Inference budget moves accuracy along d_c

$\text{max_tokens} \uparrow \cdot \text{CoT} \cdot \text{reasoner}$

5 controlled interventions, all +

reproducible · controllable · externally anchored without new human labels